DLOPs Assignment-1

Deadline **: 24/01/2026 11:59 PM**

Submission Guidelines:

1. Use Python and PyTorch for performing experiments.
2. Submit a single report with all results and the detailed analysis.
3. **Share a Colab link in the report that includes already run experiments. Without the Colab link, the student will get 0 Marks. Do not share code files separately.**
4. **Report file naming convention: Rollnumber_Name_Ass1.pdf. For example: B22CS043_Firstname_Surname_Ass1.pdf**
5. Follow all the GitHub guidelines mentioned below.

Github Guidelines:

- Create your GitHub profile.
- Create a repository named: MLOps-Name-rollNumber
- Create a branch named Assignment 1.
- In this branch, push the .ipynb file and results (one best model, train-val graphs, results, etc.) and report.
- Your readme for this assignment should contain the below mentioned tables and results.
- Create GitHub pages for your repository, which needs to be updated after each assignment.

NOTE: For training, you don't have to initialize the models with pre-trained weights.
[Optional] The tag in the document below refers to the settings you may choose not to use when performing the experiments.

Q1(a). Train a deep learning model on the MNIST and FashionMNIST datasets with 70%-10%-20% train-val-test split. Use the Following Models: ResNet-18 (use pretrained = False) and ResNet-50 (use pretrained = False). Fill in the classification accuracy in the table below and provide a detailed analysis based on the results.

			Test Classification Accuracy (in %)	
Batch Size	**Optimizers**	**Learning Rate**	**ResNet-18**	**ResNet-50**
16	**SGD**	**0.001**		
16	**SGD**	**0.0001**		
16	**Adam**	**0.001**		
16	**Adam**	**0.0001**		

32	**SGD**	**0.001**		
32 **[Optional]**	**SGD**	**0.0001**		
32 **[Optional]**	**Adam**	**0.001**		
32	**Adam**	**0.0001**		

Note: Also perform experiments by varying hyperparameters.
pin_memory=False,True
Epochs = vary the number of epochs by yourself (perform an experiment with any two total numbers of epochs).
Keep these factors Constant:- USE_AMP=True
The valid experiments include achieving a Classification accuracy of more than 80+%

Q1(b). Also train a SVM classifier on MNIST and FashionMNIST datasets with varying SVM hyperparameters with kernels such as: 'poly', 'rbf'. Report Testing Classification Accuracy with training time (in ms).

Q2. For the **FashionMNIST** dataset, report performance and train-time based on running the experiments on CPU and GPU both. Write a detailed report analyzing the training time, number of FLOPs used and the classification accuracy of the models.

				Test Classification Accuracy (in %)			**Train Time (in ms)**			**FLOPs**		
Compute	**Batch Size**	**Optimizer**	**Learning Rate**	**ResNet-18**	**ResNet-32 [Optional]**	**ResNet-50**	**ResNet-18**	**ResNet-32 [Optional]**	**ResNet-50**	**ResNet-18**	**ResNet-32 [Optional]**	**ResNet-50**
CPU	**16**	**SGD**	**0.001**									
CPU	**16**	**Adam**	**0.001**									
GPU	**16**	**SGD**	**0.001**									
GPU	**16**	**Adam**	**0.001**									